

2026 edition

Deep Learning for Music Analysis and Generation

Music Transcription

(audio $\rightarrow$ scores)

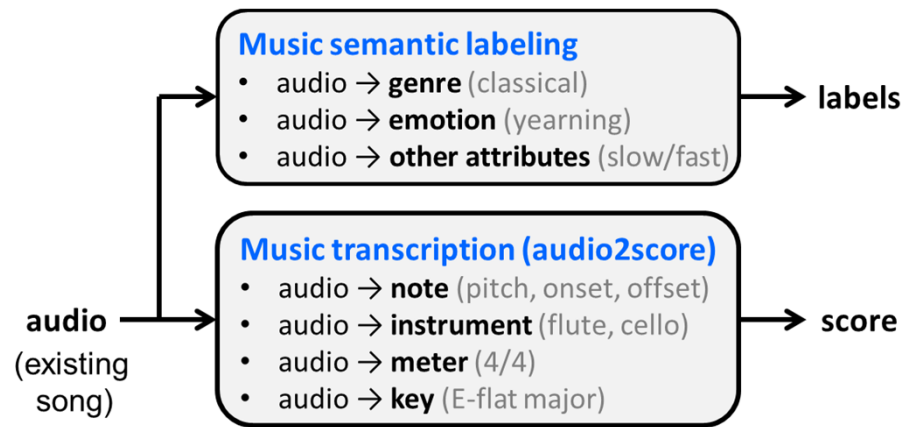


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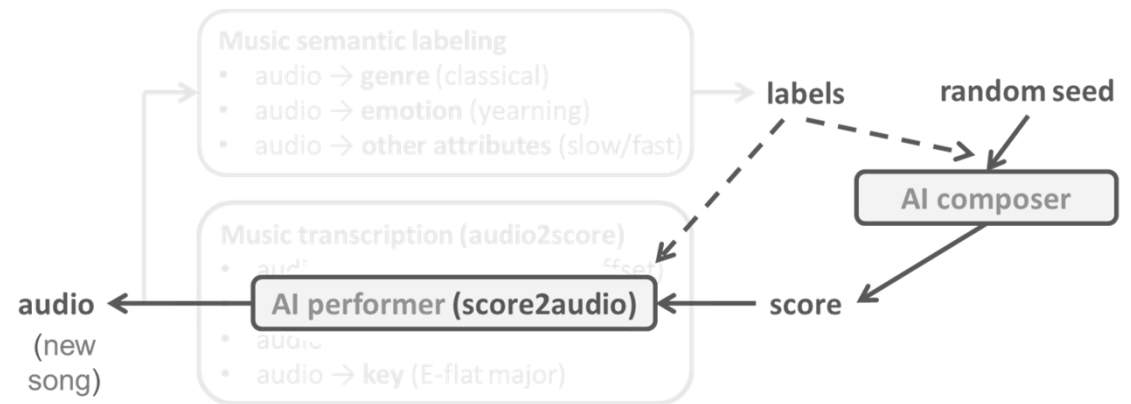
Music AI; or *Music Information Research* (MIR)

- **Music analysis**



- music understanding
- music search
- music recommendation

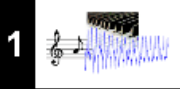
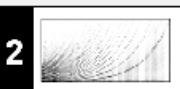
- **Music generation**

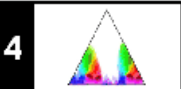
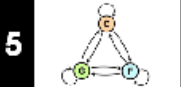
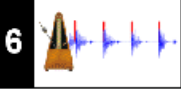
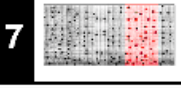
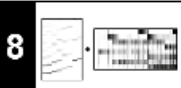


- MIDI generation
- audio generation
- MIDI-to-audio generation

Reference 1: FMP Notebook

<https://www.audiolabs-erlangen.de/resources/MIR/FMP/C1/C1.html>

Part	Title	Notions, Techniques & Algorithms	HTML	IPYNB
	Basics	Basic information on Python, Jupyter notebooks, Anaconda package management system, Python environments, visualizations, and other topics	[html]	[ipynb]
	Overview	Overview of the notebooks (https://www.audiolabs-erlangen.de/FMP)	[html]	[ipynb]
	Music Representations	Music notation, MIDI, audio signal, waveform, pitch, loudness, timbre	[html]	[ipynb]
	Fourier Analysis of Signals	Discrete/analog signal, sinusoid, exponential, Fourier transform, Fourier representation, DFT, FFT, STFT	[html]	[ipynb]
	Music Synchronization	Chroma feature, dynamic programming, dynamic time warping (DTW), alignment, user interface	[html]	[ipynb]

Part	Title	Notions, Techniques & Algorithms	HTML	IPYNB
	Music Structure Analysis	Similarity matrix, repetition, thumbnail, homogeneity, novelty, evaluation, precision, recall, F-measure, visualization, scape plot	[html]	[ipynb]
	Chord Recognition	Harmony, music theory, chords, scales, templates, hidden Markov model (HMM), evaluation	[html]	[ipynb]
	Tempo and Beat Tracking	Onset, novelty, tempo, tempogram, beat, periodicity, Fourier analysis, autocorrelation	[html]	[ipynb]
	Content-Based Audio Retrieval	Identification, fingerprint, indexing, inverted list, matching, version, cover song	[html]	[ipynb]
	Musically Informed Audio Decomposition	Harmonic/percussive separation, signal reconstruction, instantaneous frequency, fundamental frequency (F0), trajectory, nonnegative matrix factorization (NMF)	[html]	[ipynb]

Reference 2: ISMIR 2018 & 2021 Tutorials

<https://rachelbittner.weebly.com/other-resources.html>

Tutorials

Programming MIR Baselines from Scratch: Three Case Studies

2021

International Society for Music Information Retrieval (ISMIR) conference

- Part 1: **Transcription with NMF** (Ethan Manilow)
- Part 2: **Pitch Tracking with pytorch** (**Rachel Bittner**)
- Part 3: **Instrument Classification with OpenL3 & Tensorflow** (Mark Cartwright)

 See the recording here.

Fundamental Frequency Estimation in Music

2018

International Society for Music Information Retrieval (ISMIR) conference

- Part 1: **Pitch** (Alain de Cheveigné)
- Part 2: **Polyphonic fundamental frequency estimation** (**Rachel Bittner**)
- Part 3: **Applications** (Johana Devaney)

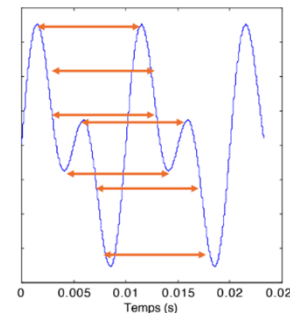
Monophonic F0 Estimation

- Part 1: **Pitch** (Alain de Cheveigné)

<https://drive.google.com/file/d/1uWCwE03dM0j0Ptp1g5vPsT56nPd0DCW2/view>

- Pitch: depends on fundamental frequency (F0), not on waveform shape
- Frequency-domain F0 analysis**
 - For example: return the frequency bin that has the highest energy per frame
 - Problems: harmonics can be stronger, missing fundamental, etc
- Time-domain F0 analysis
(based on **auto-correlation**)

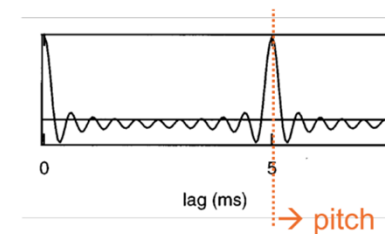
```
librosa.pyin(y, *, fmin, fmax, sr=22050,
```



$$A_t(\tau) = \sum_W x(t)x(t - \tau)$$

↑ ↑ ↑
integration window product time lag

"shift and compare"



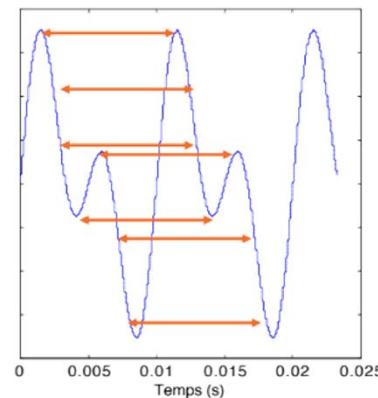
Monophonic F0 Estimation

- Part 1: **Pitch** (Alain de Cheveigné)

<https://drive.google.com/file/d/1uWCwE03dM0j0Ptp1g5vPsT56nPd0DCW2/view>

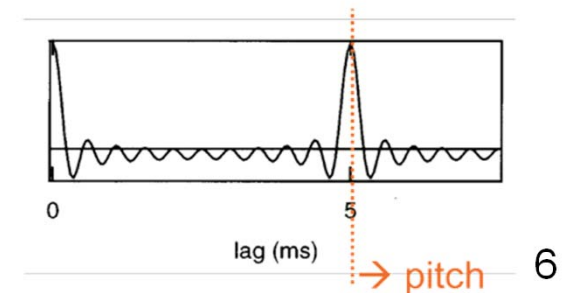
- Pitch: depends on fundamental frequency (F0), not on waveform shape
- Frequency-domain F0 analysis
- Time-domain F0 analysis** (based on **auto-correlation**)
 - YIN (2002), pYIN (2014)

```
librosa.pyin(y, *, fmin, fmax, sr=22050,
```



$$A_t(\tau) = \sum_W \overset{\text{integration window}}{x(t)x(t-\tau)} \overset{\text{product}}{\quad} \overset{\text{time lag}}{\tau}$$

"shift and compare"

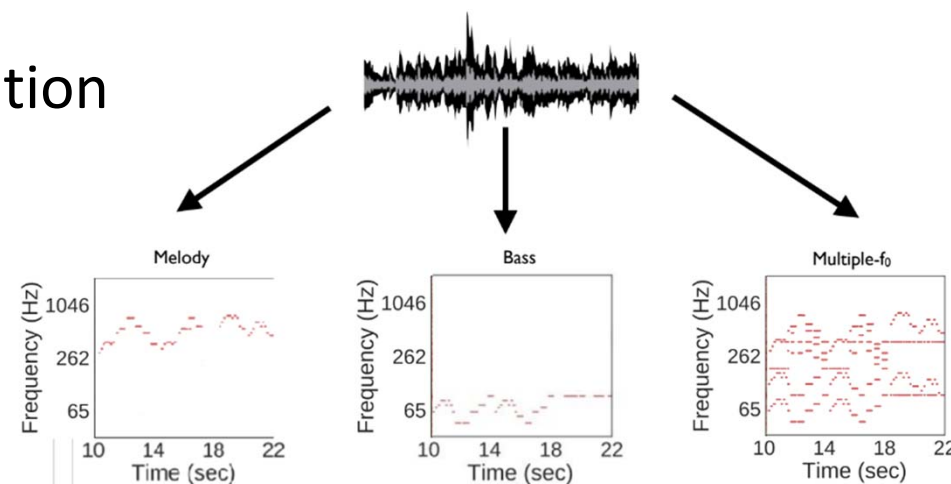


F0 Estimation

• Part 2: (Rachel Bittner)

https://drive.google.com/file/d/1Jmj6tNBGFMIEIEQldgNZ2_YgD17gS6Sv/view

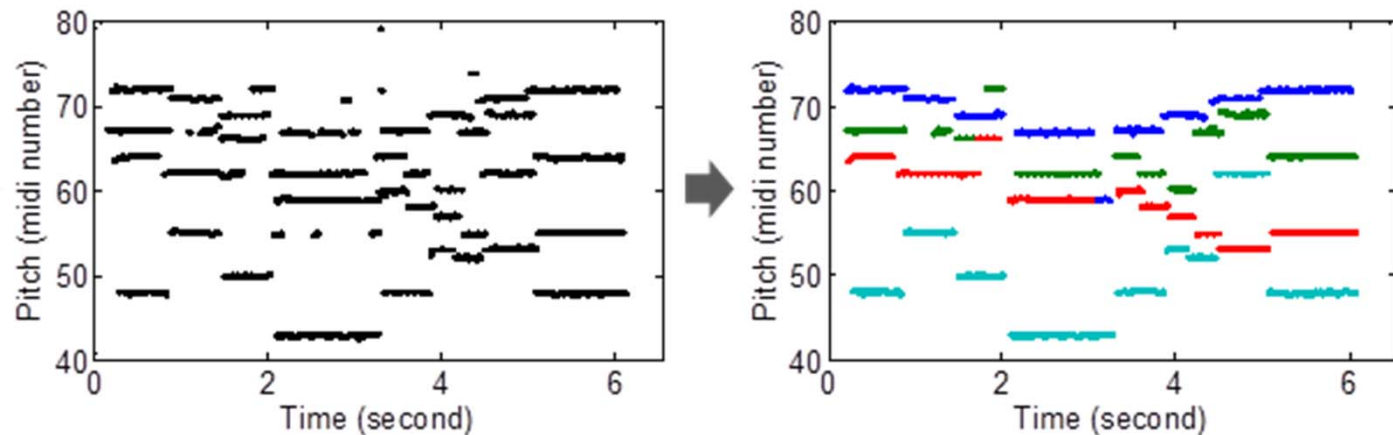
- **Monophonic** input vs **polyphonic** input
 - For example, singing solo vs. mixed song
- **Single F0 (melody)** vs **multiple F0** estimation
 - For example, vocal melody extraction
- **F0 estimation (hz)** vs **note estimation** (MIDI pitch)
- Time quantization or not
- Output instrument labels or not



F0 Estimation: Background

https://drive.google.com/file/d/1Jmj6tNBGFMIEIEQldgNZ2_YgD17gS6Sv/view

- Related tasks
 - frame-level f0 estimation
 - note estimation
 - streaming
 - score transcription
- Time resolution
- Frequency resolution
- Voicing
- Instrument labeling



Melody Extraction vs. Note Transcription

- **Melody extraction**: F0 (can reflect *overshoot*, *vibrato*, *glissando*, etc)
- **Note transcription**: MIDI pitch (*quantized in frequency*)

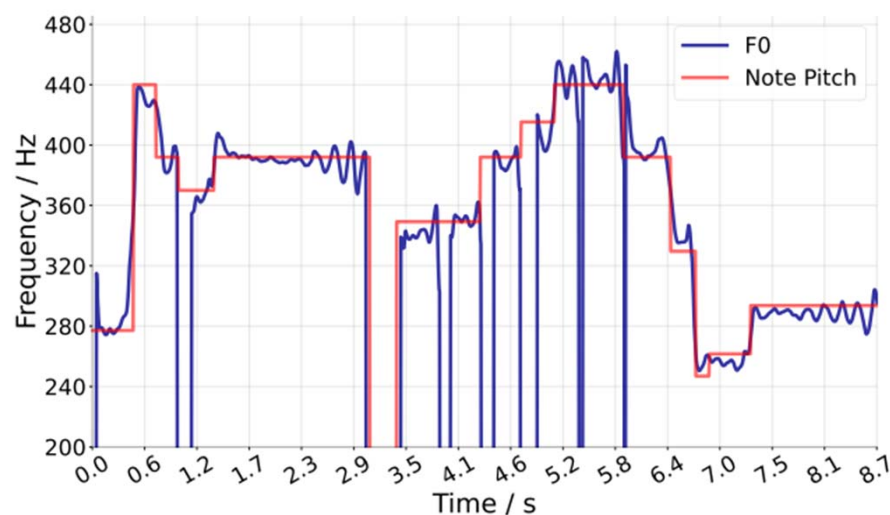


Figure source: M4Singer paper
(<https://openreview.net/pdf?id=qIDmAaG6mP>)

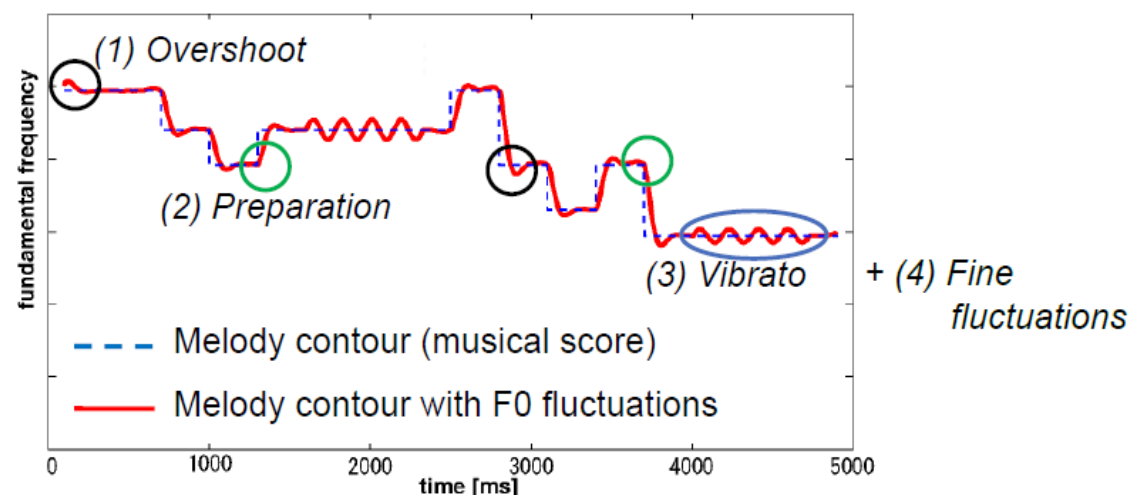


Figure from: Saitou et al, "Speech-to-singing synthesis: converting speaking voices to singing voices by controlling acoustic features unique to singing voices," WASPAA 2007

Pianoroll vs MusicXML

- **Time quantization:** quarter note, eighth note etc
- More than that...
- Usually, the target is to output a single-track or multi-track pianoroll

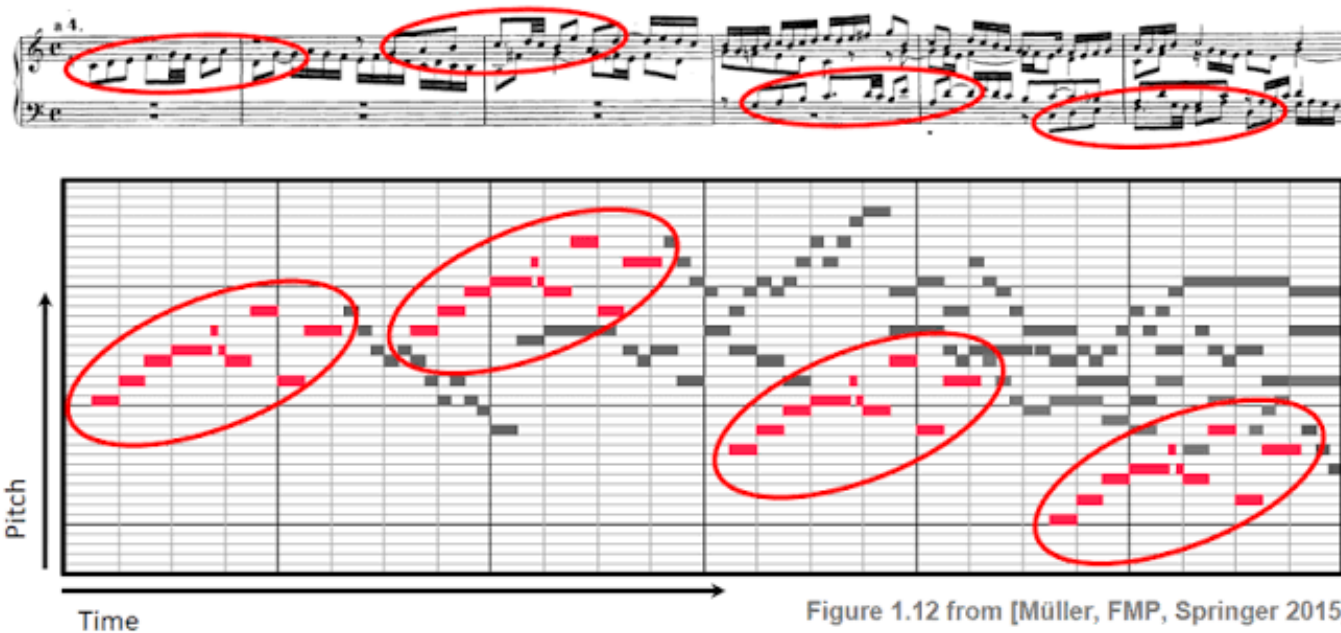
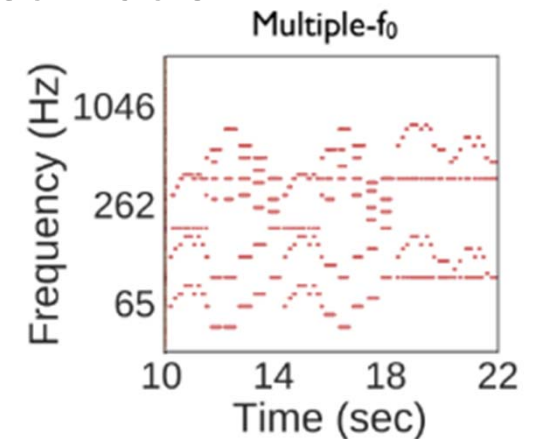
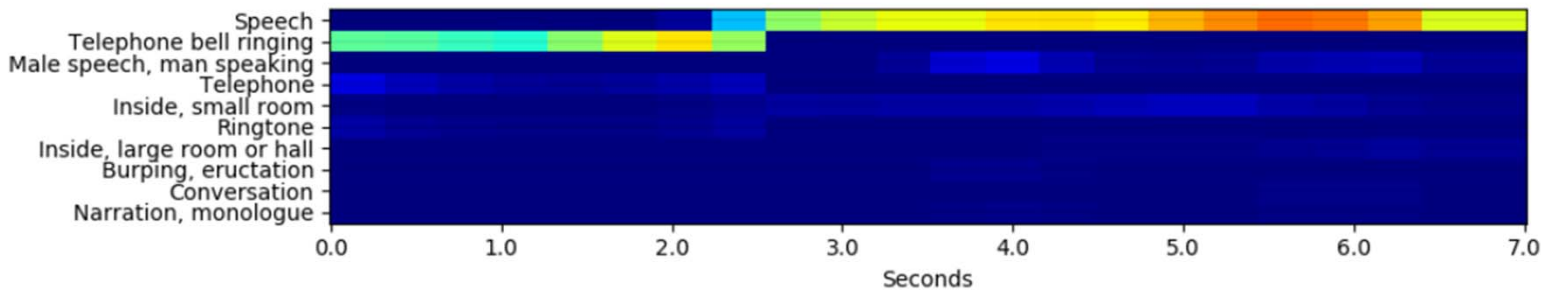


Figure 1.12 from [Müller, FMP, Springer 2015]

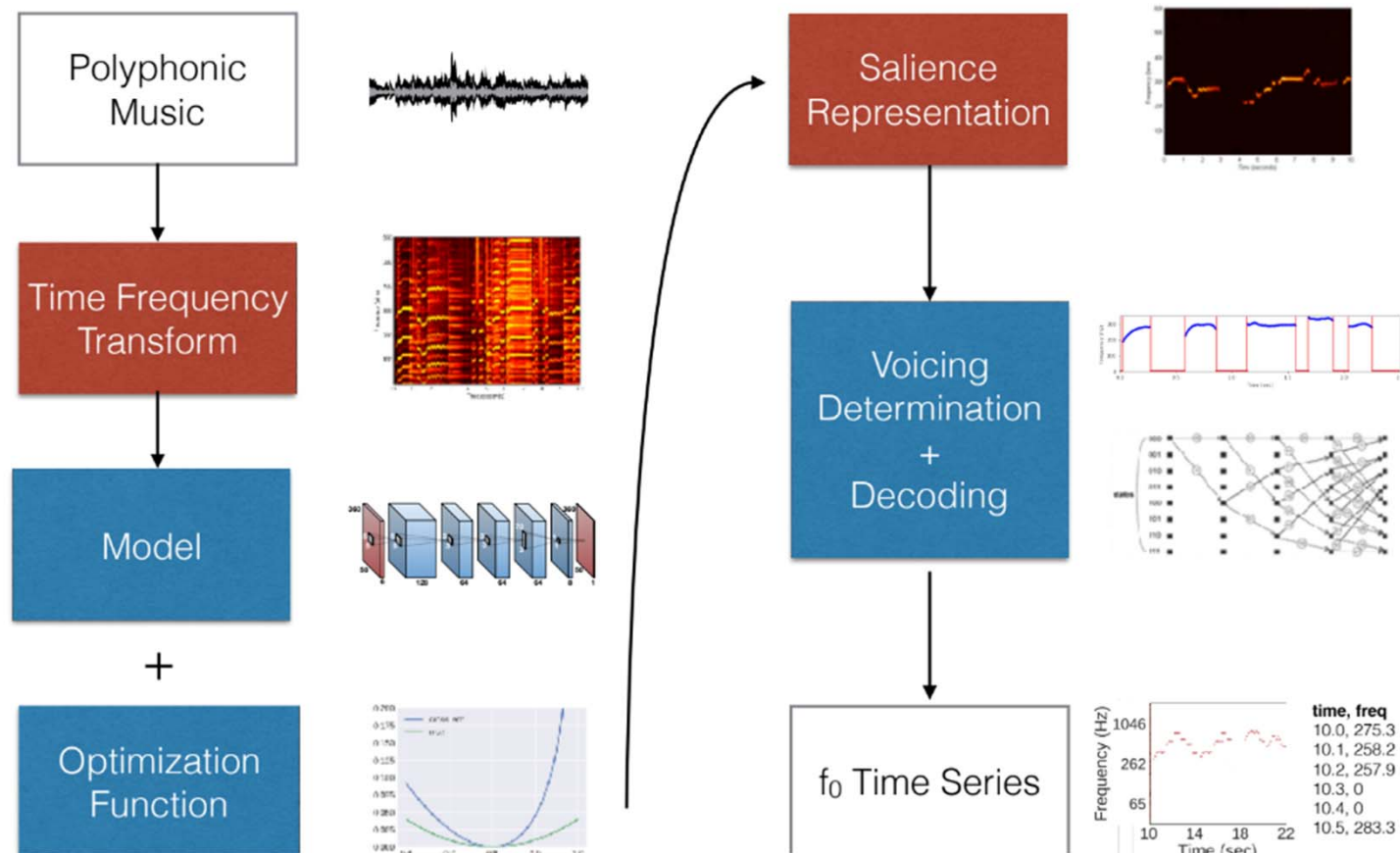
From an ML/DL Viewpoint

- **Per song:** genre classification
 - we can make predictions for each chunk and then aggregate the result over time (e.g., by taking the average logits or by majority voting on the chunk-level decisions)
- **Per short-time chunk:** audio event detection, instrument activity detection
 - e.g., output is a matrix [class x time]
- **Per time-frequency point:** f_0 estimation, multi-pitch estimation
 - e.g., output is a matrix [frequency x time]



Polyphonic F0 Estimation: Typical Approach

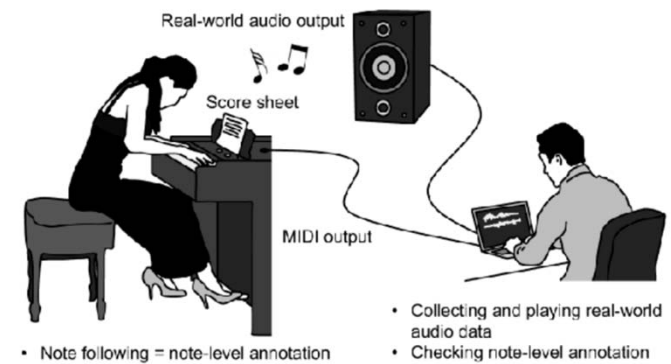
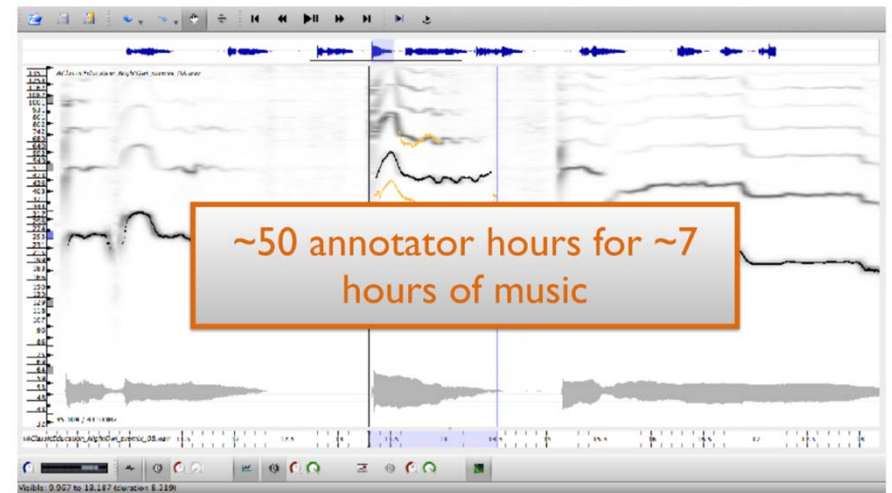
https://drive.google.com/file/d/1Jmj6tNBGFMIEEQldgNZ2_YgD17gS6Sv/view



Supervised Training

https://drive.google.com/file/d/1Jmj6tNBGFMIEIEQldgNZ2_YgD17gS6Sv/view

- But labeled data is hard to collect
- Slow down research progress
 - Single F0 estimation for mono signals is nearly solved
 - Multi-F0 estimation still has room for improvement
 - Note transcription for arbitrary instruments remains hard
 - One exception is piano note transcription, which is nearly solved, partly thanks to high-quality piano synthesizers



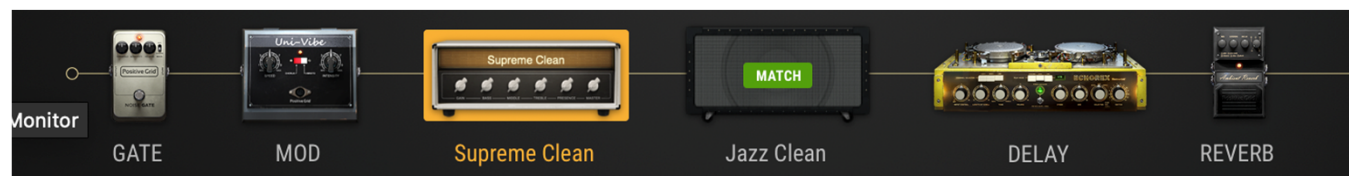
Guitar Transcription: Need to be Invariant to Audio Effects

(Examples provided by **Positive Grid®**)

Dry (single
coil EG)



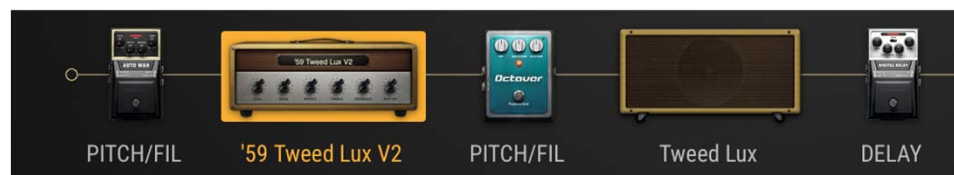
*In the
Clouds*



*Overdriven
Verb Icon*



*Lazy
Down*



Chen et al., “Towards automatic transcription of polyphonic electric guitar music: A new dataset and a multi-loss transformer model,” ICASSP 2022

Evaluation Metrics for Pitch-Related Tasks

https://craffel.github.io/mir_eval/

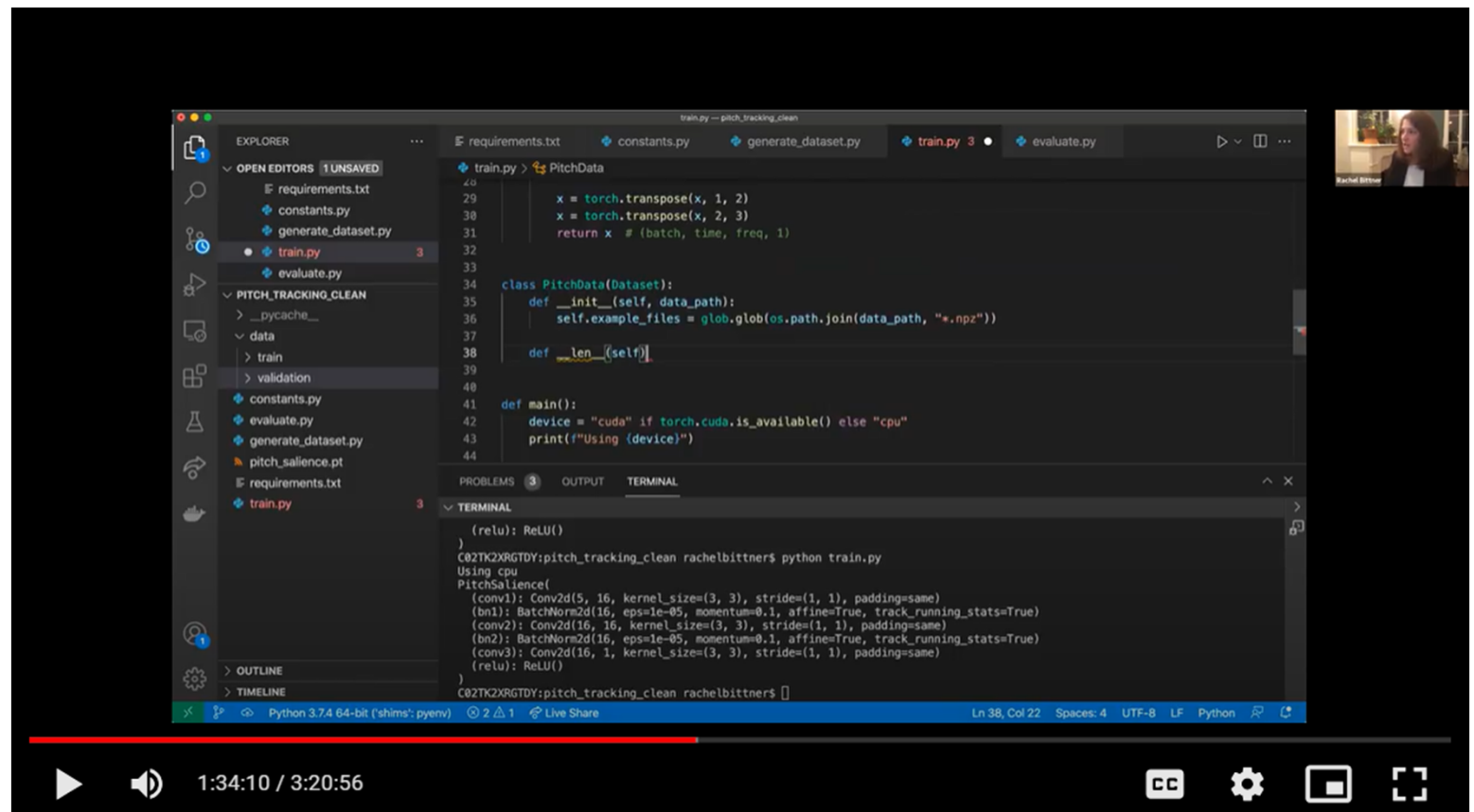
- **Single- and multi-pitch estimation**
 - correct if within 0.5 semitones of a reference frequency
 - chroma accuracy (ignore octave)
 - voicing measures
- **Piano transcription**
 - onset tolerance window: 50ms
 - offset tolerance window: 20% of note duration
 - onset only vs. onset+offset
- **Chord transcription**
 - root
 - majmin
 - majmin_inv
 - thirds
 - triads
 - tetrads
 - sevenths
 - overseg, underseg, seg

ISMIR 2021 Tutorial: Programming MIR Baselines from Scratch - Pitch Tracking

https://github.com/rabitt/ismir-2021-tutorial-case-studies/tree/main/pitch_tracking

- Video online

<https://drive.google.com/file/d/18LNaKy2ymFjEWj19gHy25wgtFV4pdML0/view>

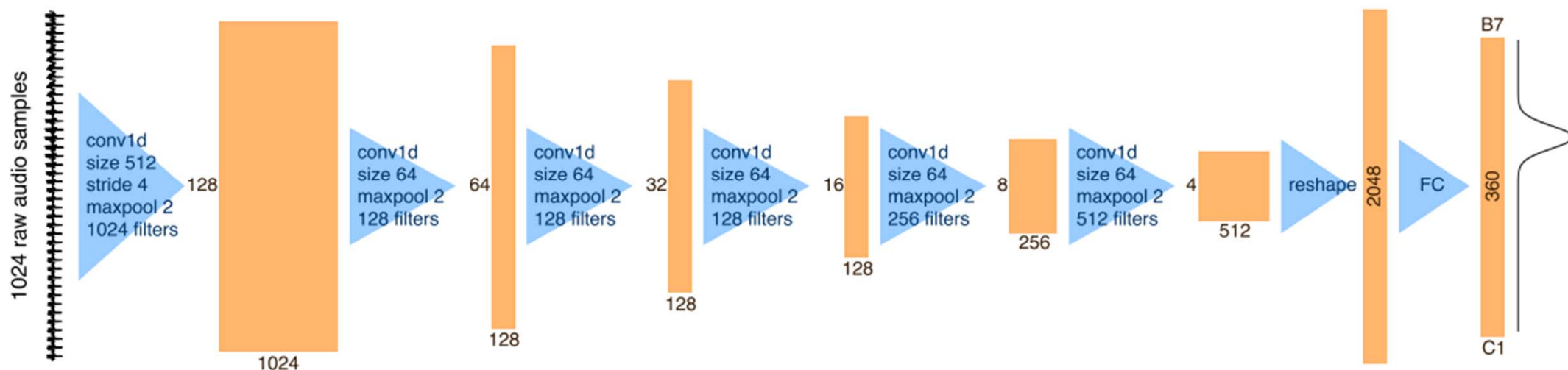


Tasks & Exemplar Models

- **Melody extraction**
 - F0 estimation in monophonic music
 - F0 estimation in polyphonic music
- **Leadsheet recognition**: quantized melody + chord
- **Multipitch estimation & multi-instrument transcription**

Melody Extraction from “Monophonic” Music: CREPE

<https://github.com/marl/crepe>



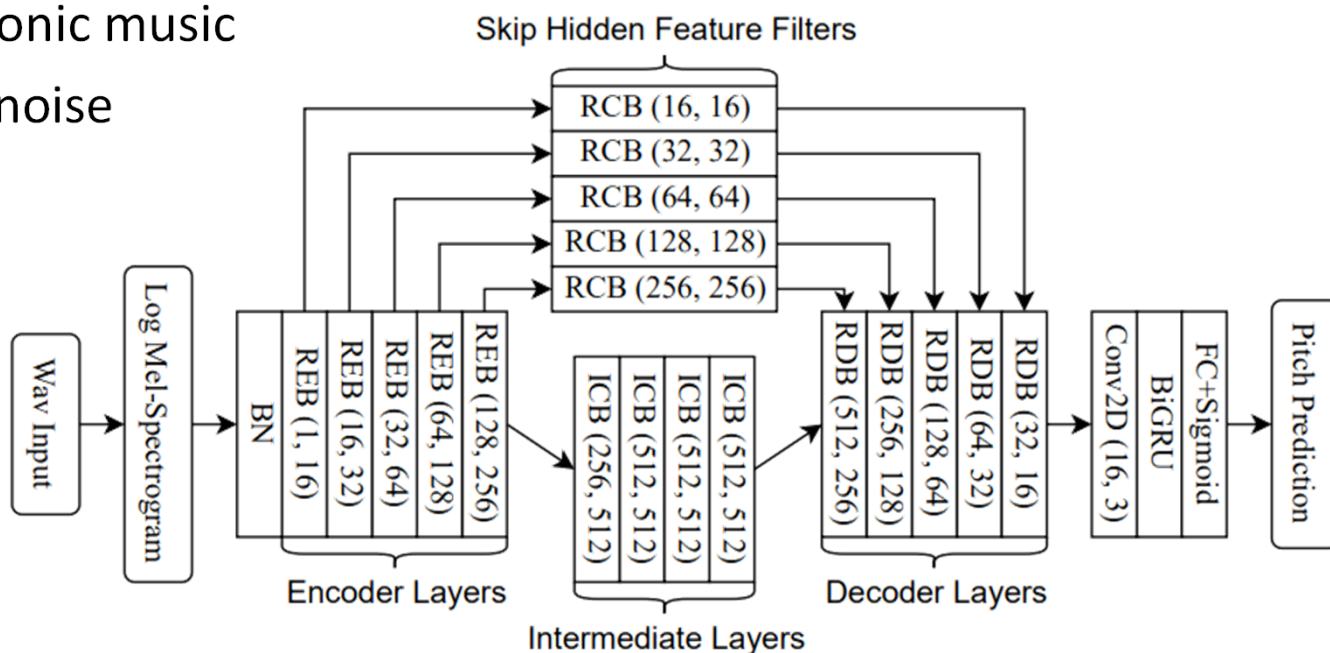
- Make a prediction every 1,024 samples
- Freq resolution: 20 cents (1/5 semitone)
- Outperform DSP-based methods such as pYIN and SWIPE

Dataset	Threshold	CREPE	pYIN	SWIPE
RWC-synth	50 cents	0.999±0.002	0.990±0.006	0.963±0.023
	25 cents	0.999±0.003	0.972±0.012	0.949±0.026
	10 cents	0.995±0.004	0.908±0.032	0.833±0.055

Vocal Melody Extraction from “Polyphonic” Music: RMVPE

<https://github.com/Dream-High/RMVPE>

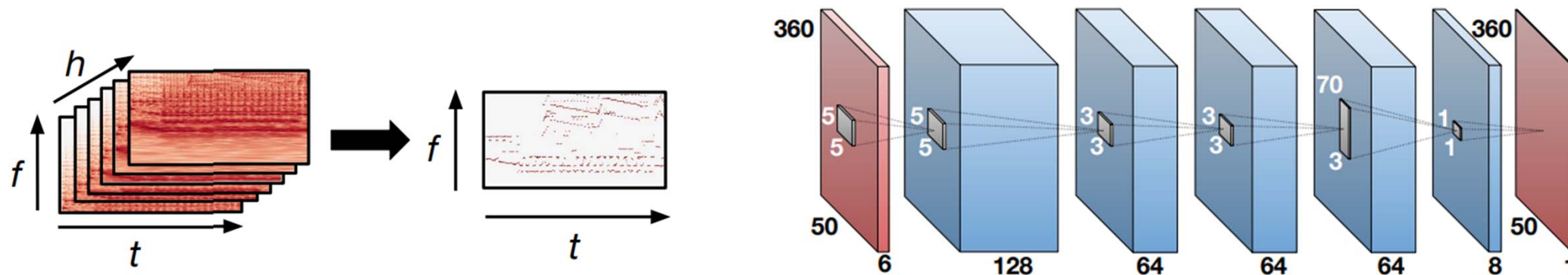
- Use deep U-Net and GRU to directly extract **vocal** f0 from polyphonic music
- Robust to different types of noise



Wei et al., “RMVPE: A robust model for vocal pitch estimation in polyphonic music,” INTERSPEECH 2023

Melody Extraction and Multi-F0 Estimation: DeepSaliency

<https://github.com/rabitt/ismir2017-deepsaliency>



- Input: harmonic constant-Q transform (HCQT) with 6 channels
- Model: 5-layer CNN (**no strides, no pooling → no shift-invariance**)
 - (5 x 5): 1 semitone in frequency and 50 ms in time
 - (70 x 3): 14 semitones in frequency to capture relationships between frequency content within an octave

Tasks & Exemplar Models

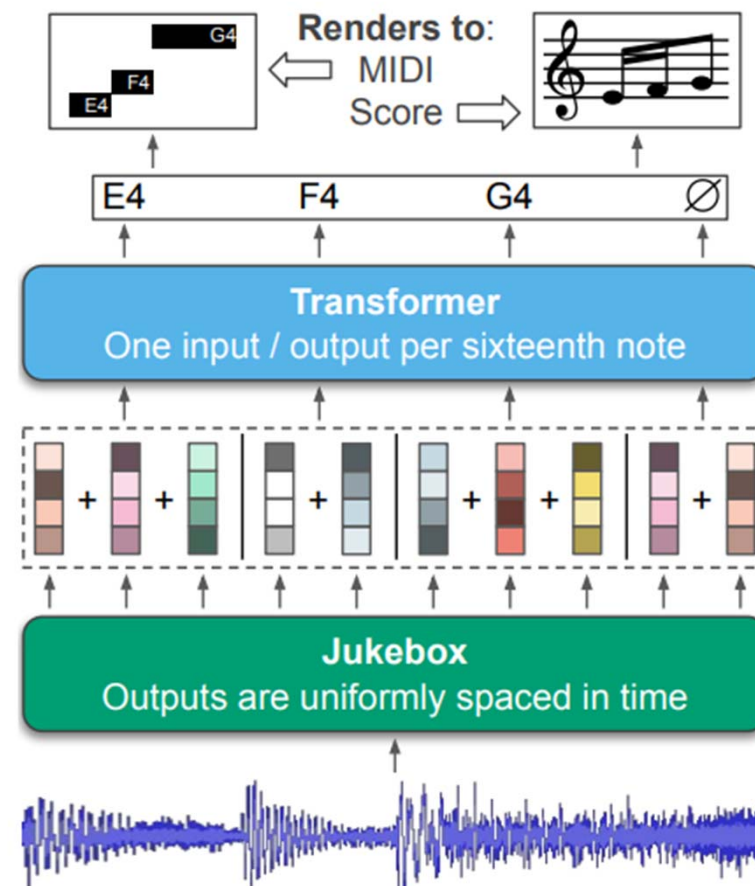
- Melody extraction
- **Leadsheet recognition:** “quantized” melody + chord
- Multipitch estimation & multi-instrument transcription

Lead Sheet Transcription: Sheet Sage

<https://github.com/chrisdonahue/sheetsage>

- Predict both melody and chord
 - Use a large pre-trained model called **Jukebox** as model backbone and then do transfer learning
 - Use **Transformer** to learn the language model (LM) for melody and chords
 - Computationally heavy but pretty accurate
 - Lighter alternatives: BTC
(<https://github.com/jayg996/BTC-ISMIR19>)

Donahue et al., “Melody transcription via generative pre-training,” ISMIR 2022



Lead Sheet Transcription: Sheet Sage-Pro

<https://github.com/pingw220/sheetsage-pro>

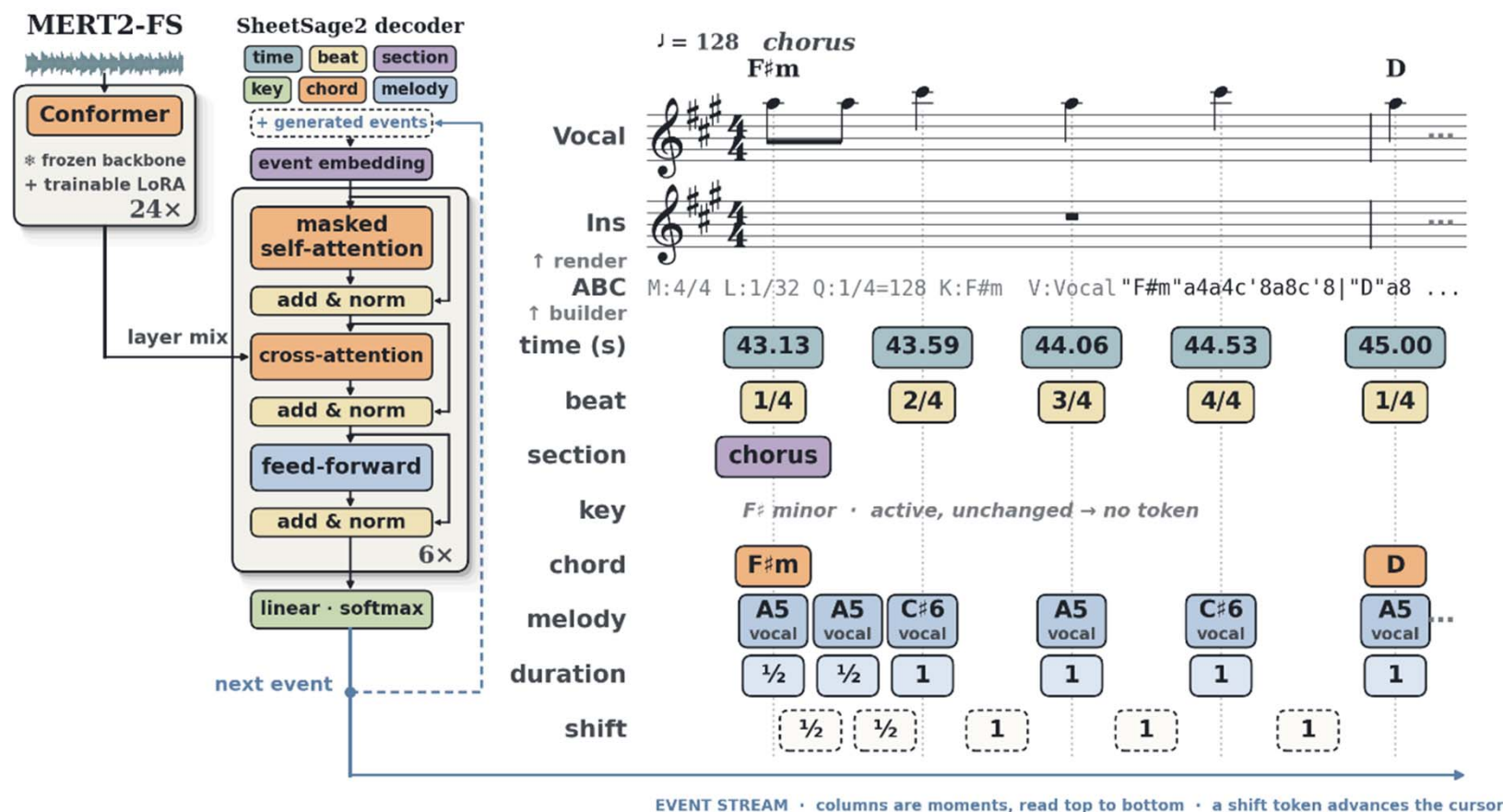
- Component-wise upgrade
 - “Mel-Band RoFormer” for vocal separation
 - “Beat This!” for beat/downbeat tracking
 - “GAME” for melody extraction
 - Jiang’s model for chord recognition
 - “All-in-One” for structure/meter analysis
- Output melody+chord+**lyrics**
 - Lyrics **with timestamps**
 - Via lyrics transcription or audio-lyrics alignment

Task	Metric	Sheet Sage	SheetSage-Pro
Melody, ZH	Note F1@50 ↑	0.049	0.250
Melody, EN	Note F1@50 ↑	0.112	0.340
ST500, ZH	Note F1@50 ↑	0.400	0.578
ST500, ZH	Onset F1@50 ↑	0.514	0.645
Chords	MajMin WCSR ↑	0.621	0.904
Chord bounds	Boundary F1 ↑	0.559	0.799
Lyrics, ZH	Syll. onset ↓	—	24 ms
Lyrics, EN	Word onset ↓	—	33 ms

Wang et al., “SheetSage-Pro: Bilingual lyric-aligned lead-sheet transcription from music audio,” ISMIR-LBD 2026

Lead Sheet Transcription: Sheet Sage2

<https://huggingface.co/m-a-p/SheetSage2>

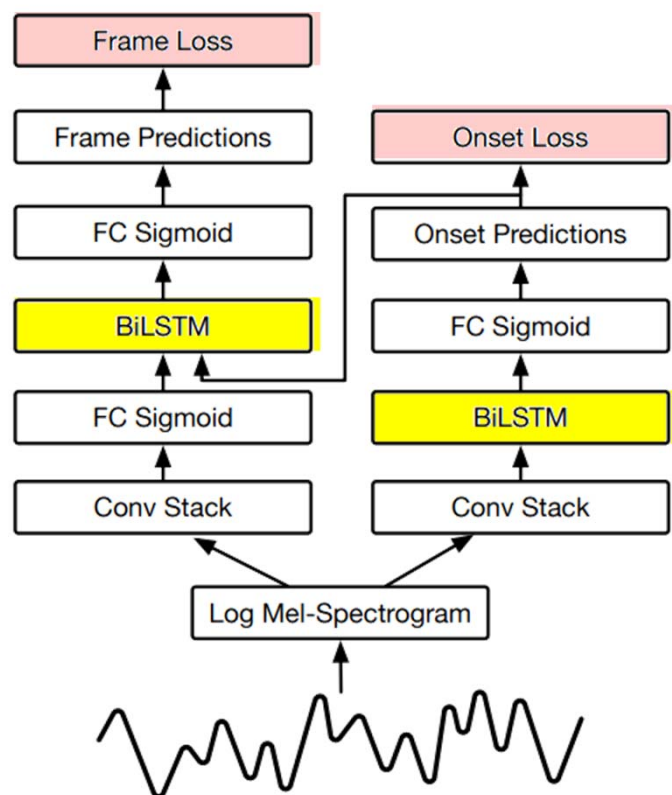


Tasks & Exemplar Models

- **Melody extraction**
- **Leadsheet recognition:** quantized melody + chord
- **Multipitch estimation & multi-instrument transcription**
 - Multipitch estimation in single instrument (piano; output is a single-track pianoroll)
 - Multipitch estimation in general music (output is a single-track pianoroll)
 - Multi-instrument transcription (output is a multi-track pianoroll)

Piano Transcription: Onset-and-Frames

<https://magenta.tensorflow.org/onsets-frames>



Frame and Onset Predictions



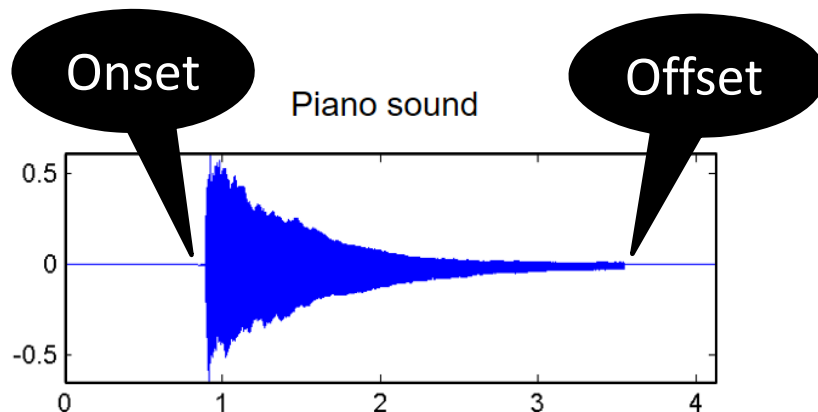
Frame Predictions Restricted by Onset Predictions



Pitch, Onset, Offset, Velocity

<https://magenta.tensorflow.org/datasets/maestro>

“We partnered with organizers of the International **Piano-e-Competition** for the raw data used in this dataset. During each installment of the competition virtuoso pianists perform on **Yamaha Disklaviers** which, in addition to being concert-quality acoustic grand pianos, utilize an integrated high-precision **MIDI capture** and playback system.”



Dynamic's note velocity			
Dynamic	Velocity*	Voice	
ppp	16	Whispering	
pp	33	Almost at a whisper	
p	49	Softer than speaking voice	
mp	64	Speaking voice	
mf	80		
f	96	Louder than speaking	
ff	112	Speaking loud	
fff	126	Yelling	



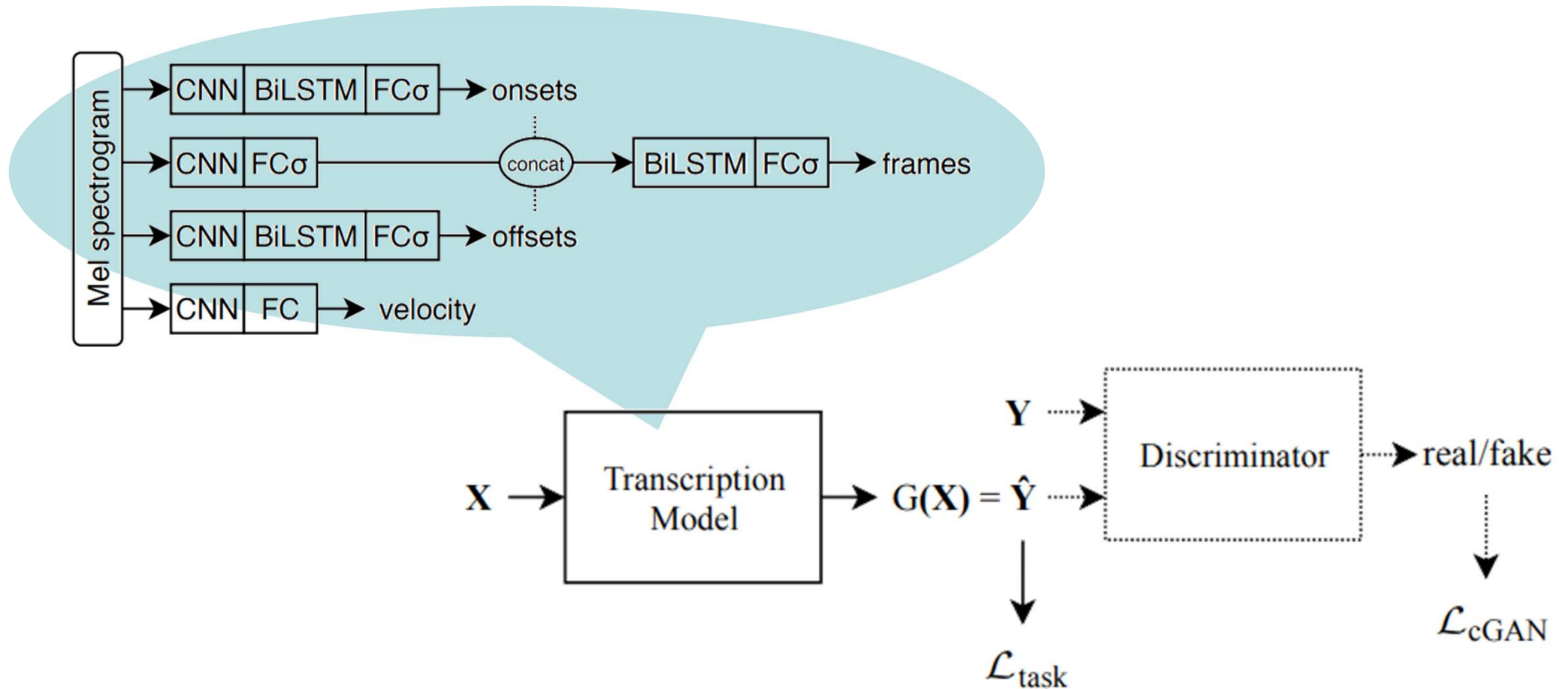
Decrescendo (diminuendo) Crescendo Accent

*Note velocity adopted from Logic Pro

<https://freeonlinesheetmusic.wordpress.com/2016/01/10/dynamics/>

Hawthorne et al., “Enabling factorized piano music modeling and generation with the MAESTRO dataset,” ICLR 2019

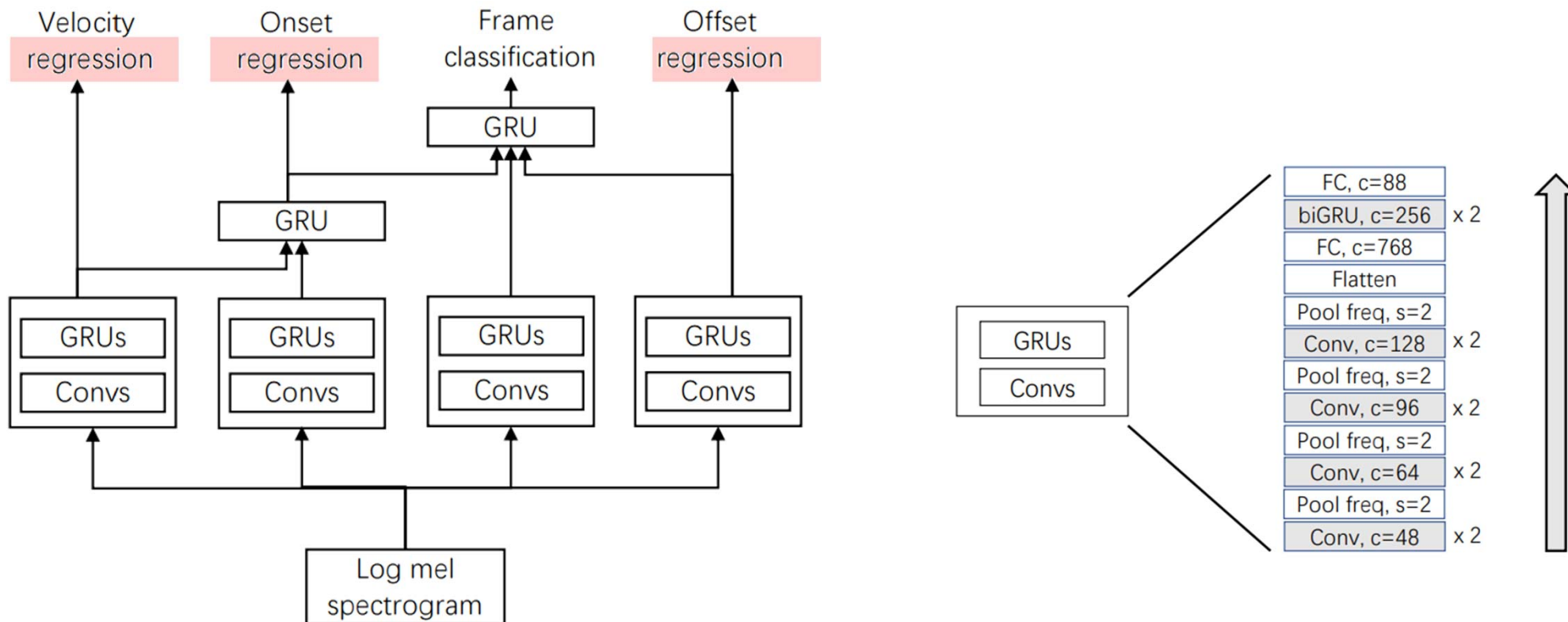
Piano Transcription with GAN



Kim & Bello, "Adversarial learning for improved onsets and frames music transcription," ISMIR 2019

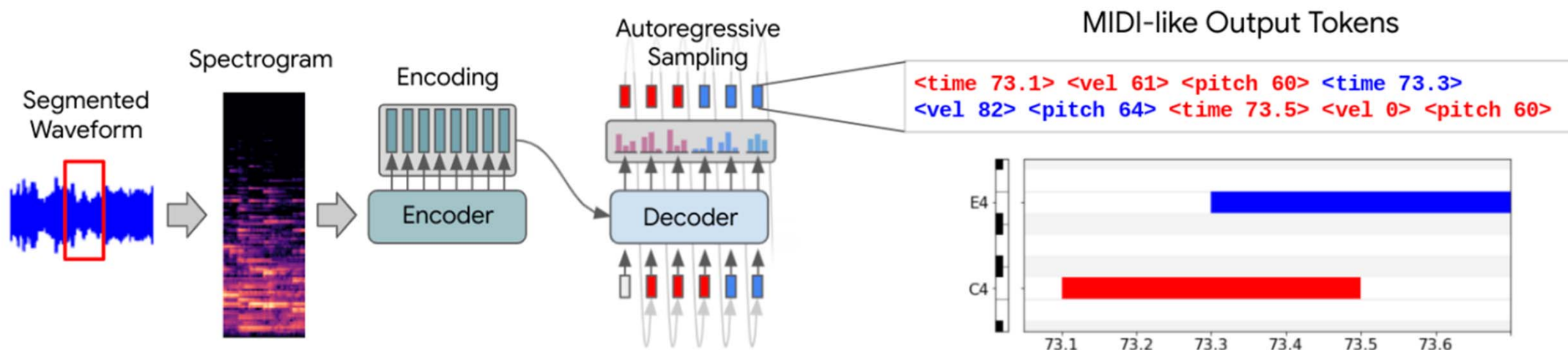
High-resolution Piano Transcription

https://github.com/bytedance/piano_transcription



Kong et al., "High-resolution piano transcription with pedals by regressing onsets and offsets times," TASLP 2021

Seq2Seq Transformers for Piano Transcription

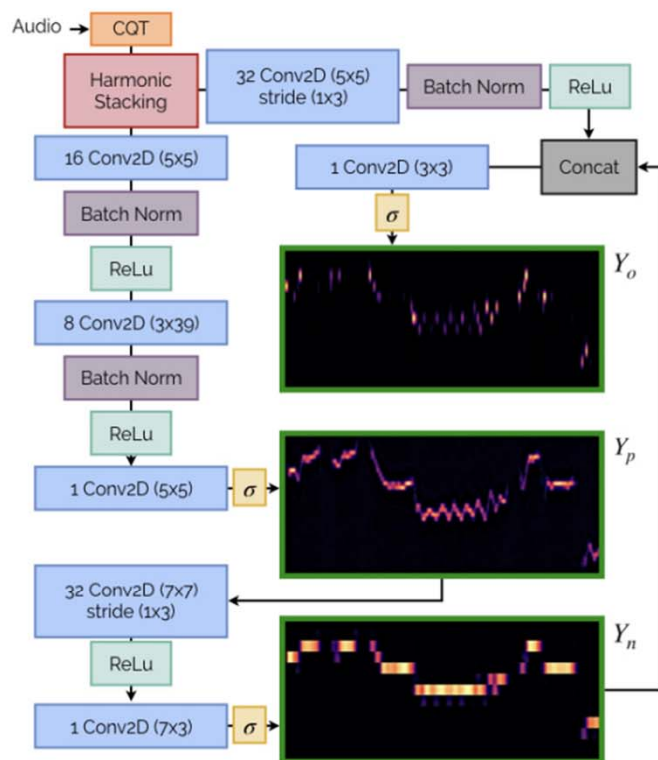


- Jointly modeling audio features and language-like output dependencies
- Possible to pre-train the decoder

Model		Onset, Offset, & Velocity F1	Onset & Offset F1	Onset F1
MAESTRO V1.0.0	Transformer (ours)	82.18	83.46	95.95
	Kong et al. 2020 [19]	80.92	82.47	96.72
	Kwon et al. 2020 [21]	—	79.36	94.67
	Kim & Bello 2019 [20]	80.20	81.30	95.60
	Hawthorne et al. 2019 [2]	77.54	80.50	95.32

Note Transcription from Polyphonic Music: Basic Pitch

<https://github.com/spotify/basic-pitch>



- For *instrument-agnostic* note transcription and multi-pitch estimation
- Pure CNN-based for being light-weight

Try Basic Pitch, a free audio-to-MIDI converter with pitch bend detection, built by Spotify. [Learn more](#) or follow the instructions below.

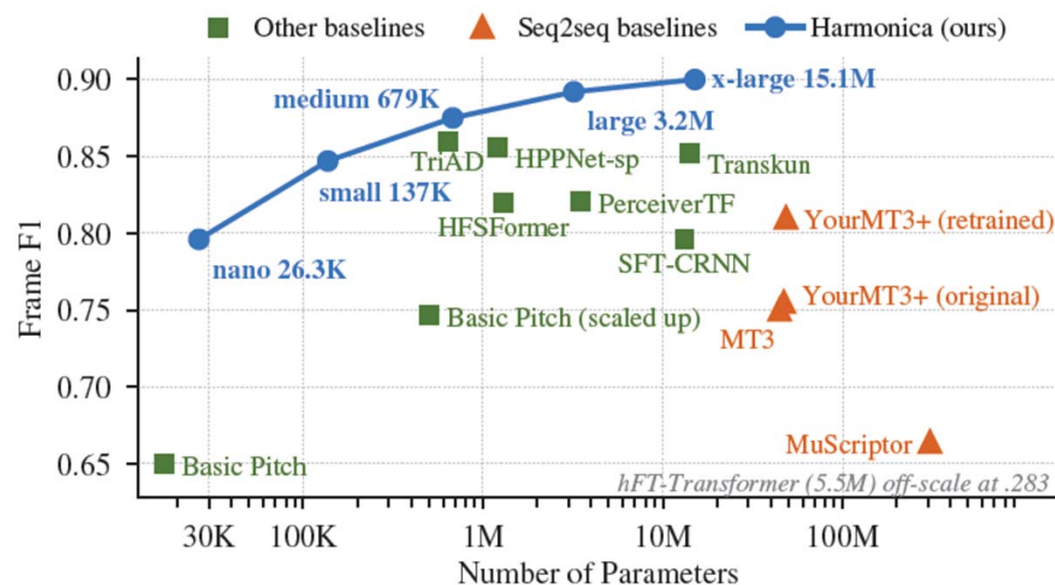
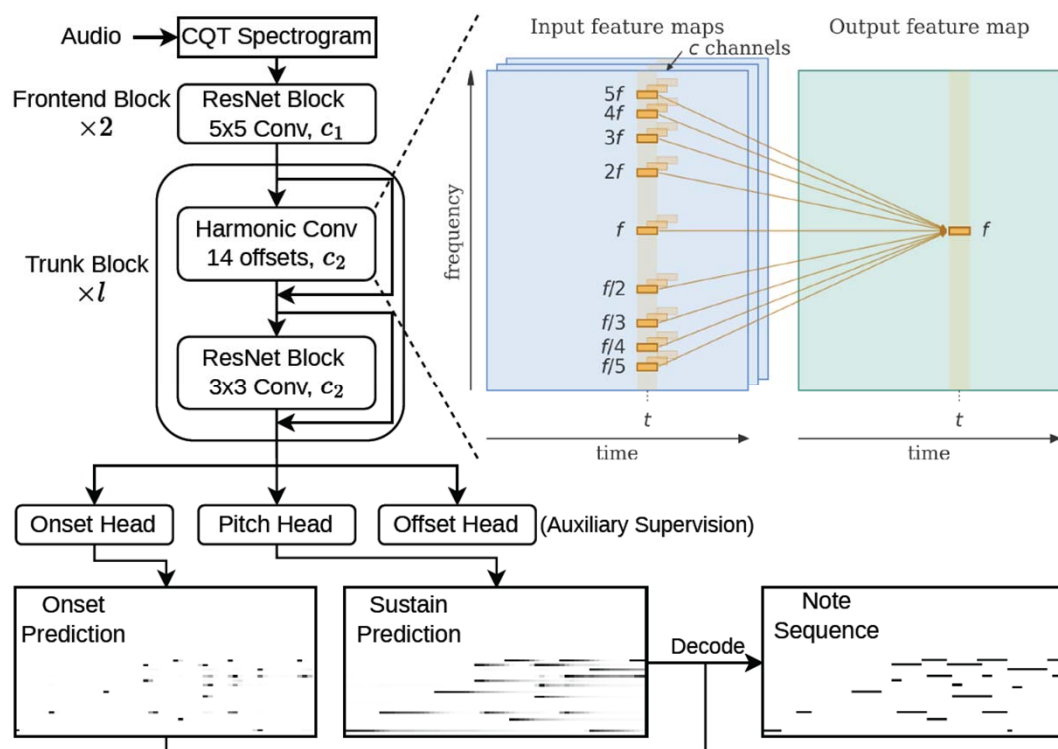
- 1 — Press record and sing a ditty into your computer. Or drop a recording of any single instrument (piano, guitar, xylophone, you name it).
- 2 — Then get a MIDI version back. Just like that.
- 3 — Download the MIDI file to fine tune and make corrections in your favorite digital audio workstation.

Fig. 1. The NMP architecture. The matrix posterioriagram outputs Y_o , Y_p , and Y_n are outlined in green. σ indicates a sigmoid activation.

Bittner et al., “A lightweight instrument-agnostic model for polyphonic note transcription and multipitch estimation,” ICASSP 2022

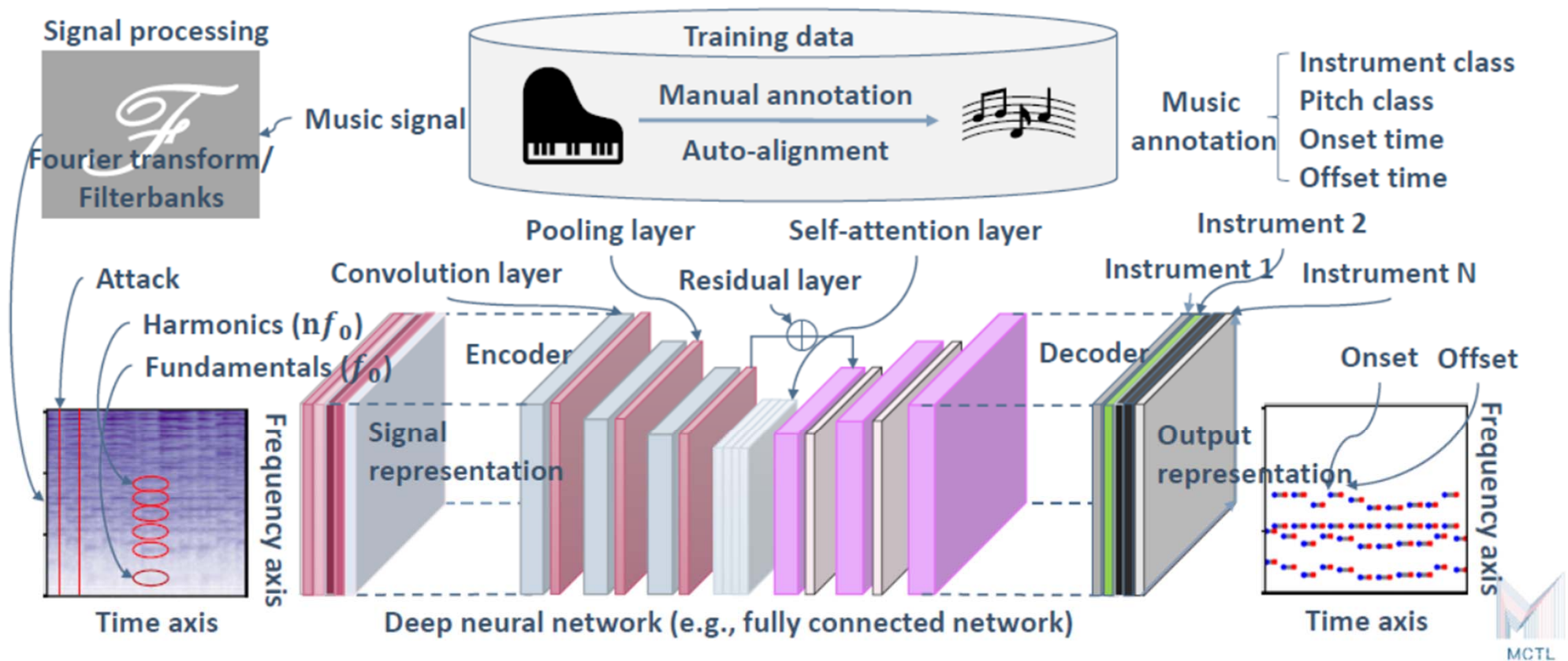
Note Transcription from Polyphonic Music: Harmonica

<https://www.oulongshen.xyz/amt>



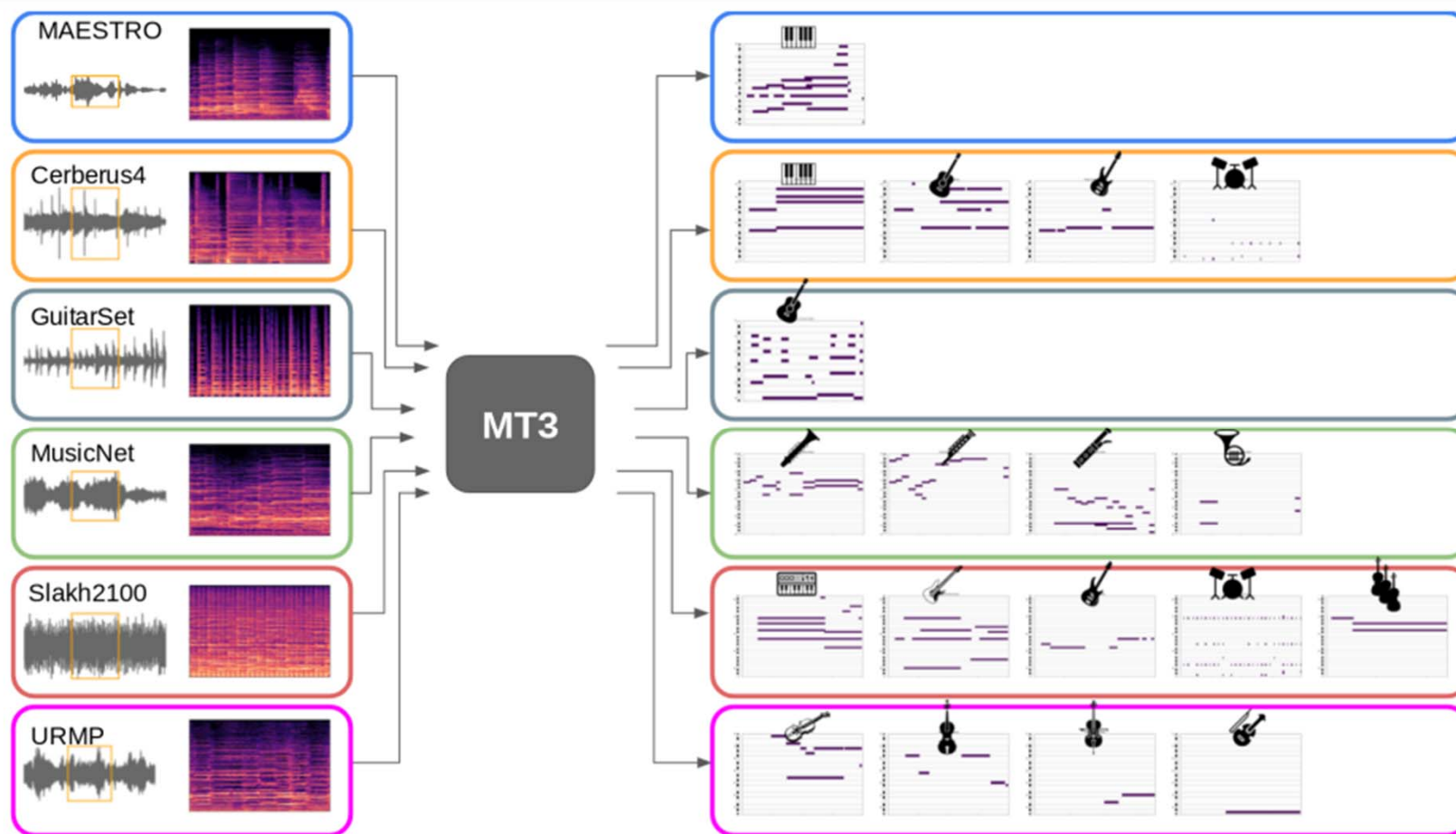
Multi-Instrument Transcription: Omnizart

<https://github.com/Music-and-Culture-Technology-Lab/omnizart>



Wu et al., "Omnizart: A general toolbox for automatic music transcription," JOOS 2021

Multi-Instrument Transcription: MT3



Multi-Instrument Transcription: MuseScriptor

<https://github.com/muscriptor/muscriptor>

- SOTA model trained with 170k recordings (11k hours) with aligned note annotations, alongside a synthetic dataset of 1.45M MIDI files
- Excellent result when instrument tags are given

